

COMPUTER COMMUNICATION NETWORKS

Major Project Assignment
Network Design & Simulation using GNS3

Total Marks: 100 | Duration: 4 Weeks | Team Size: 1-5 Students

Course Code	CSE248	Semester	Spring 2026
Instructor	Tasbiha, Waseem, Zain	Submission	Week 1 + Week 4

1. Project Overview

This project is a core component of the Computer Communication Networks (CCN) course. Students will design, configure, and simulate a complete computer network using GNS3 (Graphical Network Simulator-3) running on an Ubuntu virtual machine. The project is structured into two phases: an initial proposal & setup submission at the end of Week 1, and a final submission at the end of Week 4.

The objective is to provide students with hands-on exposure to real-world network engineering concepts including IP addressing, routing protocols, VLANs, network services, and documentation practices — all within a simulated environment.

2. Learning Outcomes

Upon successful completion of this project, students will be able to:

1. Install and configure GNS3 on an Ubuntu virtual machine environment.
2. Design a multi-device network topology with proper IP addressing schemes.
3. Configure routing protocols (static/RIP/OSPF/EIGRP) and inter-VLAN routing.
4. Deploy and test network services such as DHCP, DNS, NAT, and HTTP.
5. Analyze and troubleshoot network connectivity using packet tracing tools.
6. Produce professional network documentation including topology diagrams and configuration reports.

3. Project Timeline

Week	Phase	Tasks	Deliverable
------	-------	-------	-------------

Week 1	Setup & Proposal	Install Ubuntu VM + GNS3, select network type, submit proposal report	Phase 1
Week 2	Design & Build	Build topology in GNS3, assign IPs, configure routers and switches	<i>Internal</i>
Week 3	Testing & Documentation	Configure services, test connectivity, prepare final report and demo video	<i>Internal</i>
Week 4	Final Submission	Submit full package: demo video, final report, GNS3 project files	Phase 2

4. Phase 1 — Setup & Proposal (End of Week 1)

Ph1 VM Setup + Network Proposal Report

End of Week 1

20 Marks

4.1 VM & GNS3 Setup — Live Demo

Each team must demonstrate a working GNS3 environment to the instructor during or before the end of Week 1. The demo must show:

- Ubuntu 22.04 LTS (or 24.04) running inside a hypervisor (VirtualBox / VMware / KVM).
- GNS3 version 2.2.x or later installed and launched successfully.
- At least one router image (e.g., Cisco IOS or Cisco IOSv) imported and bootable in GNS3.
- At least one simple test topology (two routers connected) running without errors.
- Internet connectivity from inside the VM (ping 8.8.8.8).

4.2 Proposal Report Requirements

Alongside the demo, each team must submit a typed proposal document (PDF, minimum 4 pages) covering:

Required Sections in Proposal Report

1. Title Page — Project title, team members (names + roll numbers), date
2. Selected Network Type — with justification (see Section 5 for options)
3. Network Topology Diagram — hand-drawn or tool-generated (Cisco Packet Tracer / draw.io)
4. IP Addressing Plan — subnets, device IPs, masks, gateway assignments
5. Services to be Implemented — list of protocols and services planned
6. Division of Work — which team member is responsible for which part
7. References — at least 3 relevant sources (RFCs, textbooks, or documentation)

5. Network Type Options

Teams must select exactly one network type from the list below. The choice must be finalized in the Phase 1 proposal and cannot be changed without written approval from the instructor.

#	Network Type	Minimum Requirements
1	Small Office / Home Office (SOHO) (1 Member)	1 router, 1 switch, 4 PCs, DHCP, NAT, basic firewall rules
2	Enterprise LAN with VLANs (2 Members)	2 routers, 2 managed switches, VLANs, inter-VLAN routing (Router-on-a-Stick), DHCP per VLAN
3	Multi-Site WAN (OSPF/RIP) (2-3 Members)	3+ routers, 2+ sites, dynamic routing (OSPF or RIP), static routes, ping across WAN
4	ISP / BGP Simulation (4-5 Members)	3 autonomous systems, eBGP peering, route advertisement, NAT at edge
5	Data Center Network (3-5 Members)	Core/Aggregation/Access layers, redundant links, STP/RSTP, LACP, server VLANs

6. Restrictions & Academic Integrity Rules

⚠ READ CAREFULLY — Violation of any restriction below will result in ZERO marks for the entire project.

- Each team must work independently. Sharing GNS3 project files, configurations, or reports between teams is strictly prohibited.
- The network topology in the final submission must match exactly what was approved in the Phase 1 proposal. Any major change requires written approval.
- Pre-built or downloaded GNS3 lab templates from the internet are NOT allowed. The network must be built from scratch.
- AI-generated reports or topology designs submitted as your own work constitute academic dishonesty.
- All team members must be able to explain any part of the submitted work during the demo session.
- Late submissions will be penalized: -10% per day, up to a maximum of 3 days. After 3 days, no submission will be accepted.

6.1 Technical Restrictions

- GNS3 must be run locally on your Ubuntu VM — cloud-hosted GNS3 is not accepted.
- Cisco IOS router images are recommended. If using another vendor (MikroTik, Juniper), approval must be obtained from the instructor by the end of Week 1.
- You may use Cisco IOSv, IOS 15.x, or compatible images. Do NOT use Packet Tracer — it is a separate tool and does not fulfill the GNS3 requirement.
- All device configurations must be saved and included in the submission ZIP archive.

6.2 Report Restrictions

- Reports must be submitted in PDF format only. Word/DOCX files will not be accepted.
 - Minimum page count: 4 pages (Proposal) and 10 pages (Final Report), excluding cover page and references.
 - All topology diagrams must be clearly labeled with device names, interface IDs, and IP addresses.
 - Plagiarism (copying from internet sources without citation) will result in an automatic F on the assignment.
-

7. Phase 2 — Final Submission (End of Week 4)

Ph2	Demo Video + Final Report + GNS3 Project Files	End of Week 4	80 Marks
-----	--	---------------	----------

7.1 Demo Video

Each team must submit a screen-recorded demo video (10–15 minutes) demonstrating the fully working network. The video must show:

- **GNS3 opened with the full topology loaded and all devices running.**
 - All router/switch console windows visible.
 - No errors or stopped devices in the topology.
- **Live ping tests between all major segments of the network.**
 - PC-to-PC across VLANs / subnets.
 - PC to router gateway.
 - End-to-end across WAN links (if applicable).
- **Demonstration of at least two configured services (e.g., DHCP lease + NAT + DNS).**
- **Running show commands on routers/switches:**
 - show ip route, show ip interface brief, show running-config
- **Team member verbally explaining the network during the demo (voice narration required).**

Video Format Requirements

Format: MP4 (H.264) preferred, or MKV / AVI
Resolution: Minimum 720p (1280x720), 1080p recommended
File size: Maximum 2 GB
Audio: Voice narration is mandatory — silent videos will lose 10 marks
Filename: TeamName_RollNumbers_CCN_Demo.mp4

7.2 Final Report

The final report must be a comprehensive professional document in PDF format covering all aspects of the implemented network. Required sections:

Sec	Section Title	Key Contents
1	Executive Summary	1-page overview of the project
2	Network Design	Final topology diagram, design rationale, device list
3	IP Addressing Table	Full table of all IPs, masks, gateways per interface
4	Device Configurations	Complete running-config for all routers and switches
5	Routing Protocol Analysis	Routing tables, protocol chosen, convergence results
6	Services Configuration	DHCP, DNS, NAT, or other services with screenshots
7	Testing & Verification	Ping results, traceroute, packet captures (if any)
8	Challenges Faced	Problems encountered and how they were resolved
9	Conclusion	Summary, lessons learned, possible improvements
10	References	At least 5 sources (RFCs, books, documentation)

7.3 GNS3 Project Files

The complete GNS3 project folder must be zipped and submitted. It must include:

- The .gns3 project file.
- All startup configuration files for every device.
- Any custom scripts or automation files used.
- A README.txt file explaining the folder structure and how to open the project.

8. Submission Instructions

How to Submit

All files must be uploaded to the course LMS (Moodle / Google Classroom — as specified by instructor).

Phase 1 Submission (End of Week 1):

- File 1: TeamName_Phase1_ProposalReport.pdf
- Attend in-person demo session scheduled by instructor.

Phase 2 Submission (End of Week 4):

- File 1: TeamName_FinalReport.pdf

- File 2: TeamName_RollNumbers_CCN_Demo.mp4
- File 3: TeamName_GNS3_Project.zip (containing all project files)

Naming Convention: Use your team's name and roll numbers in all filenames.

All three files must be submitted together in a single upload. Partial submissions will NOT be graded.

9. Marking Rubric

9.1 Phase 1 Rubric (20 Marks)

Criteria	Marks	Obtained
Working Ubuntu VM with GNS3 Installed & Running	7	
Router Image Loaded and Basic Topology Demo	3	
Network Type Selected with Clear Justification	3	
Topology Diagram (Logical & Physical)	3	
IP Addressing Plan (correct subnetting)	2	
Division of Work & References in Report	2	
PHASE 1 TOTAL	20	

9.2 Phase 2 Rubric (80 Marks)

Criteria	Marks	Obtained
Demo Video — Quality, Completeness, Narration	20	
Network Fully Functional (all devices reachable)	20	
Routing Protocol Correctly Configured & Working	10	
Services Configured (DHCP, NAT, DNS, etc.)	10	
Final Report — Structure, Completeness, Diagrams	10	
IP Addressing Accuracy & Subnetting Correctness	5	
GNS3 Project Files — Organized & Openable	3	
Challenges & Reflection Section Quality	2	
PHASE 2 TOTAL	80	

GRAND TOTAL (Phase 1 + Phase 2)**100**

10. Tips for Success

- Start the VM setup in the very first week — GNS3 installation can be tricky and may require troubleshooting.
- Use VirtualBox or VMware with at least 4 GB RAM allocated to the Ubuntu VM. 8 GB recommended for complex topologies.
- Save your GNS3 project frequently. Use snapshots in your hypervisor before making major changes.
- Document as you go — write your report incrementally instead of leaving it all to Week 4.
- Record the demo video in one sitting with all devices running. Restarting devices mid-video looks unpolished.
- Test your ping connectivity between all devices before recording the video.
- Use Cisco command reference (Cisco IOS documentation) when configuring routing protocols.